



COMPUTER VISION FINAL LAB

Computer Vision Final Lab Part 1



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1 Introduction

The aim of this project is to explore and understand a traditional image classification pipeline using a Bag of Visual Words (BoVW) approach. This method transforms image classification into a task related to text-based document classification, where images are represented as collections of visual words that contain key features from the images [2]. The main steps include feature extraction using SIFT (Scale-Invariant Feature Transform) and ORB (Oriented FAST and Rotated BRIEF) [5, 6], clustering the extracted features into visual words using K-Means clustering, and encoding images as histograms of visual words. Finally, Support Vector Machines (SVMs) are used with different hyperparameters for classification, followed by an evaluation based on Mean Average Precision (mAP). In this report, the entire pipeline, the various hyperparameters tested, and the quantitative and qualitative evaluation of the results are documented. The results are presented both qualitatively and quantitatively.

2 Methods

2.1 Data Preprocessing

The dataset used for this project is CIFAR-10, consisting of 60,000 32x32 color images in 10 classes, with 6,000 images per class. For these experiments, the focus is on five classes: frog, automobile, bird, cat, and deer. The dataset is divided into training and testing sets, with 5,000 training images and 1,000 test images. The images are resized to 128x128 pixels and the pixel values are normalized between -1 and 1. This is essential for consistent feature extraction using methods such as SIFT and ORB.

2.2 Feature Extraction

2.2.1 SIFT (Scale-Invariant Feature Transform)

SIFT detects keypoints in an image that are invariant to scale, rotation, and illumination changes [5]. These keypoints represent important local features of the image, such as corners and edges. For each keypoint, SIFT extracts a feature descriptor based on the local image gradient. This descriptor is robust to small changes in viewpoint or scale, making it ideal for visual matching tasks.

2.2.2 ORB (Oriented FAST and Rotated BRIEF)

ORB is a faster alternative to SIFT. It uses the FAST (Features from Accelerated Segment Test) keypoint detector and the BRIEF (Binary Robust Independent Elementary Features) descriptor [6]. ORB is computationally less expensive and suitable for real-time applications, though slightly less robust than SIFT in detecting complex features. In this project, SIFT and ORB are used to extract keypoints and descriptors from the training images. The number of keypoints (nfeatures) is varied during experimentation to study how this affects the performance of the classifier.

2.3 Building the Visual Vocabulary (K-Means Clustering)

After extracting descriptors from all the images, they are clustered into a predefined number of clusters (visual words) using K-Means clustering [2]. Each cluster center represents a visual word in this Bag of Visual Words model. The number of visual words is varied during hyperparameter tuning to find the optimal setup for classification.

K-Means Algorithm: K-Means is an iterative clustering algorithm that partitions the data into K clusters by minimizing the variance within each cluster [2]. In this case, descriptors from images are grouped into visual words based on their similarity. Once the clusters are formed, a visual vocabulary that will be used for encoding the images is obtained.

2.4 Image Encoding (Histogram of Visual Words)

Once the visual vocabulary is established, each image is represented as a histogram of visual words. This is analogous to representing a document as a bag of words in natural language processing. For each image, the number of descriptors belonging to each visual word (cluster) is calculated by finding the nearest cluster center (using K-Nearest Neighbors) [2]. This results in a feature vector for each image that describes its distribution of visual words. This histogram-based encoding converts images into fixed-length vectors, making them suitable for input to machine learning classifiers like Support Vector Machines (SVMs).

2.5 Support Vector Machine (SVM) Classification

For the classification task, One-vs-Rest (OvR) SVMs with a linear kernel are used. The OvR approach allows us to train a separate binary classifier for each class, determining whether an image belongs to that class or not. SVM is a supervised learning algorithm widely used for classification tasks. It aims to find a hyperplane that maximally separates data points of different classes [4].

2.6 Hyperparameter Search

To find the optimal settings for the classifier, an extensive hyperparameter search is performed. The parameters experimented include:

- The number of visual words (500, 1000).
- The number of keypoints per image (20, 30).
- Subset percentages of the dataset for training the visual vocabulary (30%, 40%).
- SVM hyperparameters: kernel (rbf), C (0.1, 1), and gamma (scale, 0.01) [3].

The hyperparameter search is conducted using GridSearchCV, which performs a cartesian product of all possible parameter combinations [2]. The performance of each combination is evaluated based on the Mean Average Precision (mAP) metric. This allows us to systematically explore the hyperparameter space and find the best configuration for both the SIFT and ORB pipelines.

2.7 Evaluation Metrics

For the quantitative evaluation, the mean Average Precision (mAP), a standard metric in object detection and retrieval tasks, is calculated. The mAP reflects the average precision of the classifier across all classes, providing a single score that summarizes its overall performance. For the qualitative evaluation, the top-5 and bottom-5 ranked images for each class, based on the classifier's confidence scores, are visualized. This helps to understand how well the classifier is able to rank images of a given class, and where it makes mistakes.

3 Results and Discussion

3.1 Keypoint Detection and Feature Extraction

The parameters configured for SIFT and ORB were:

Table 1: Parameters for SIFT and ORB

Method	Parameter	Value
SIFT	nfeatures	30
SIFT	edgeThreshold	8
ORB	nfeatures	30
ORB	edgeThreshold	9

This selection allows for a sufficient number of features to enable effective classification while minimizing the processing load, particularly important given the dataset's size and complexity. The edgeThreshold parameter was tailored to allow better visualization of the detected keypoints, making the features extracted more clear.

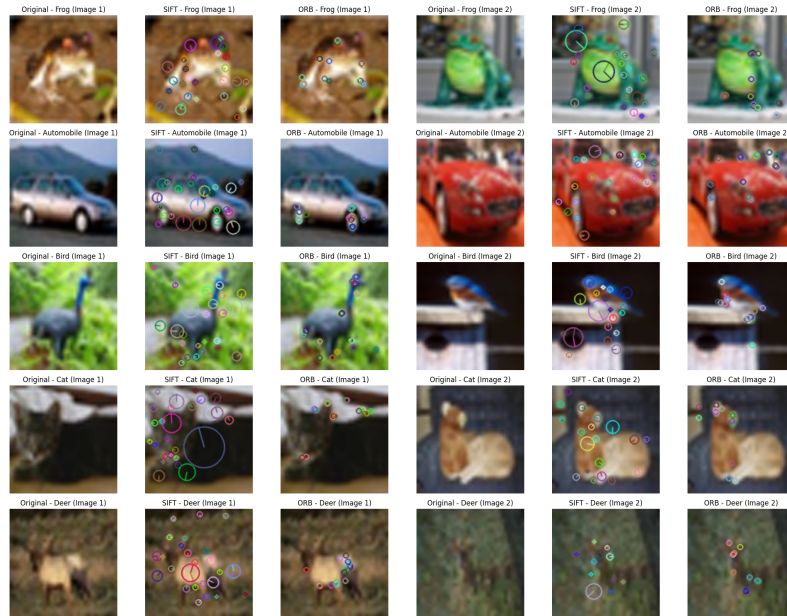


Figure 1: Keypoints detected using SIFT and ORB for various classes.

The visual representation of keypoints indicates a good distribution across the images, showing how both methods captured different aspects of the objects. However, SIFT's greater density in keypoint detection suggests it may be more reliable for complex images where feature detail is essential.

3.2 Building the Visual Vocabulary

The visual vocabulary was constructed using K-Means clustering, with the 1000 visual words, by partitioning the extracted feature descriptors. To ensure consistency with feature extraction, the same parameters were retained for clustering. This choice allows to us capture enough features for effective classification while ensuring computational efficiency.



Figure 2: K-Means clustering results illustrating the formation of visual words.

The clustering results indicated that the visual words formed a rich vocabulary from the extracted features. The clusters reflected coherent groupings, where similar features were aggregated together, indicating effective feature extraction. Using a 30% subset provided initial insights but limited the reliability of cluster centroids. Increasing to 40% resulted in more stable and representative clusters. The 50% subset yielded the richest vocabulary and improved the classification, although it increased computational costs. Thus, a balance between subset size and computational efficiency is crucial for optimizing classification performance.

3.3 Visualizing the Bag of Visual Words for Each Class

The shapes of the histograms are as follows:

Dataset	Shape
SIFT Training Histograms	(5000, 1000)
ORB Training Histograms	(5000, 1000)
SIFT Test Histograms	(1000, 1000)
ORB Test Histograms	(1000, 1000)

Table 2: Shapes of the histograms generated for SIFT and ORB on both training and test datasets.

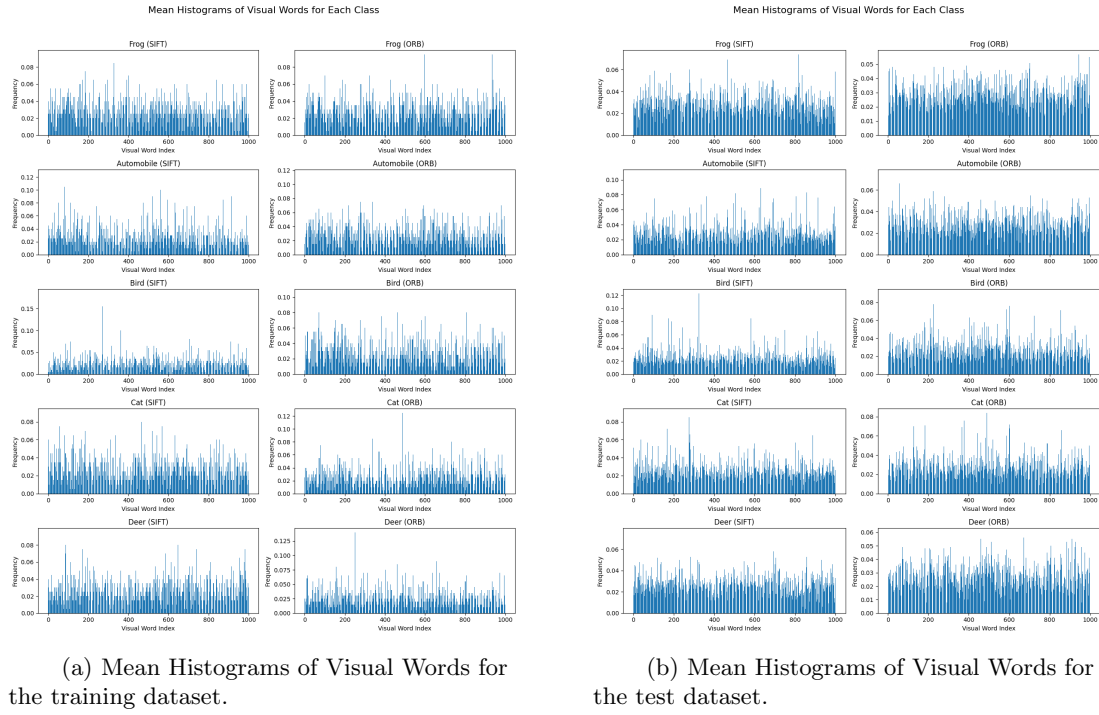


Figure 3: Comparison of Mean Histograms of Visual Words for Training and Test Datasets.

The histogram shapes demonstrate the number of features extracted and their distribution across the visual vocabulary. The training histograms reflect how effectively the classifiers captured relevant visual patterns. The non-sparse nature of the histograms indicates that both classifiers utilized a wide range of visual words, which is beneficial for the learning process. This suggests that our classifiers have access to a rich representation of the training data, allowing them to generalize better to the test dataset.

3.4 Quantitative Evaluation: Mean Average Precision (mAP)

After training, the mAP scores obtained for the test set are as follows:

Method	mAP	AP per Class
SIFT	31.8%	[28.11%, 46.8%, 30.5%, 26.3%, 27.4%]
ORB	25.1%	[25.6%, 27.2%, 25.5%, 23.2%, 24.0%]

Table 3: Mean Average Precision (mAP) scores and Average Precision (AP) per class for SIFT and ORB, evaluated on the test set.

The results indicate that SIFT outperformed ORB in overall classification accuracy, achieving higher mAP scores across most classes. The notable disparity in performance between SIFT and ORB indicates that SIFT extracted more discriminative features, particularly for the 'Automobile' class, where the mAP of 46.4% highlights its effectiveness in recognizing distinct patterns that characterize vehicles. This performance can be attributed to the more distinguished features of automobiles compared to the other classes (animals), which share more similar characteristics and therefore pose greater challenges for classification.

3.5 Qualitative evaluation: Top-5 and Bottom-5 Ranked Images

The visualizations of the top-5 and bottom-5 ranked test images for each class based on classifier confidence scores are as follows:

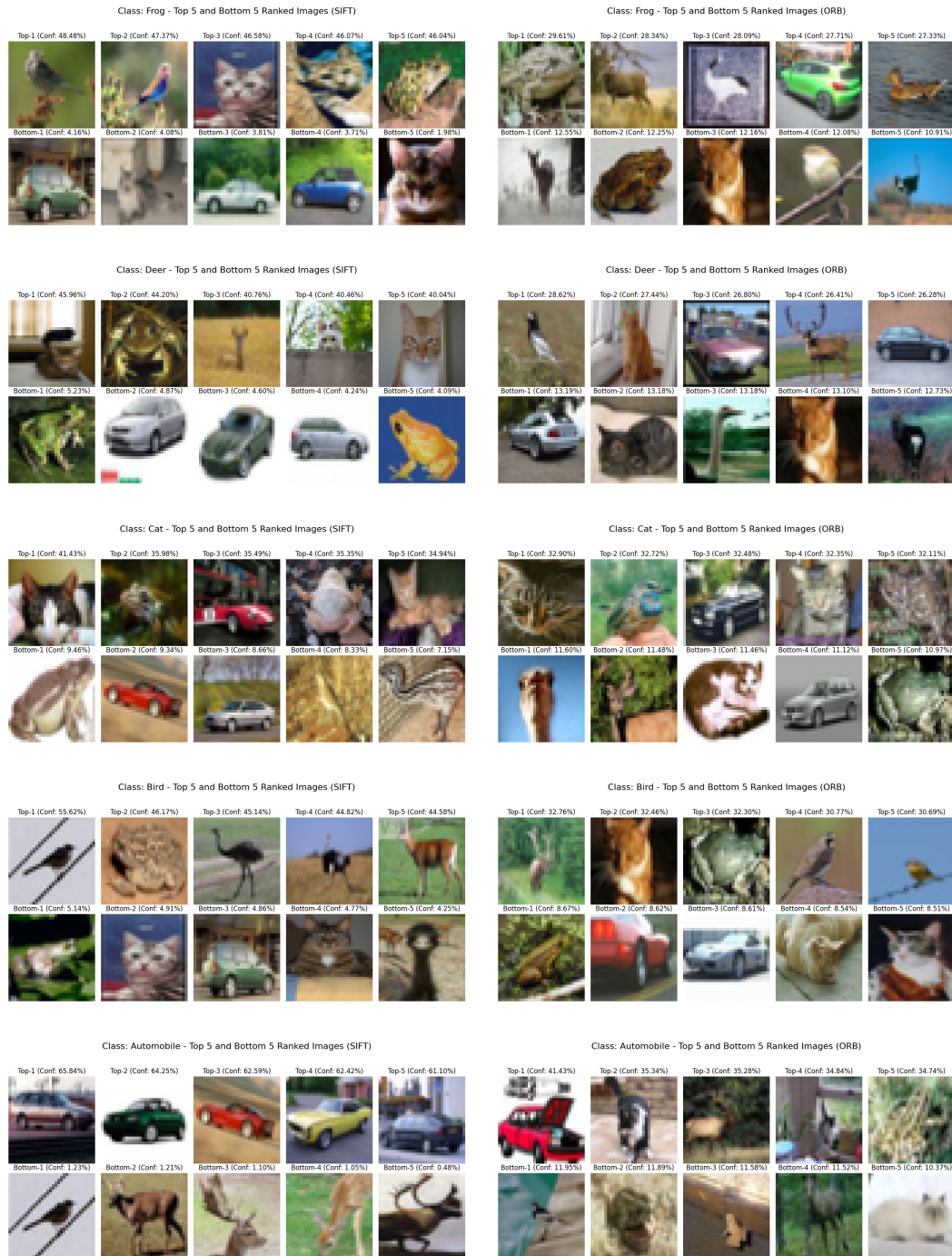


Figure 4: Comparison of Top 5 and Bottom 5 Ranked test Images for Each Class using SIFT and ORB

This analysis provides insight into the strengths and weaknesses of the classifiers. These results reveal that SIFT consistently ranked images more accurately than ORB, especially in classes with distinct features. While the top-ranked images demonstrated high confidence, suggesting effective identification of key characteristics of each class, the bottom-ranked images revealed instances where the classifier faced difficulties. For example, misclassifications often happen in images with overlapping characteristics or when background clutter complicated object detection [1]. Moreover, while some top-ranked images for certain classes may not be all correctly classified, the next ranked images might still provide good classifications. This observation sug-

gests the possibility of features being recognized even if the most confident predictions aren't as strong.

3.6 Hyperparameter Search

An extensive hyperparameter search was conducted to optimize the performance of both classifiers, focusing on the test set evaluation. The parameters experimented with are summarized below:

Parameter	Values
Number of Visual Words	500, 1000
Number of Keypoints per Image	20, 30
Subset Percentages	30%, 40%
SVM Hyperparameters	Kernel: rbf, C: 0.1, 1; Gamma: scale, 0.01

Table 4: Summary of hyperparameter settings explored during the search process.

These parameters were chosen for the grid search for computational efficiency and because they performed better in previous runs. The grid search proved effective, as evidenced by an increase in the mAP scores for both classifiers, reflecting the benefits of systematic hyperparameter tuning [3].

Method	SIFT	ORB
Best mAP Score	48.5%	42.1%

Table 5: Best Mean Average Precision (mAP) scores for SIFT and ORB, evaluated on the test set.

Parameter	SIFT	ORB
C	0.1	0.1
Gamma	'scale'	'scale'
Kernel	'rbf'	'rbf'
Number of Clusters	500	500
Subset Percentage	30%	30%
Number of Keypoints	30	20

Table 6: Best parameters for SIFT and ORB.

Results indicate that the SIFT method, while more computationally intensive, demonstrated superior accuracy. The grid search identified the optimal hyperparameters for SIFT and ORB based on their ability to balance flexibility, complexity, and computational efficiency, which are crucial for image classification tasks. A regularization parameter C of 0.1 allows for a more flexible decision boundary, helping to manage noise and improve generalization on unseen data. Setting gamma to 'scale' adjusts the influence of training samples, enabling the model to capture non-linear relationships in the data. The radial basis function (rbf) kernel facilitates the creation of complex decision boundaries, making the classifier able to differentiate between visually similar classes. Using 500 clusters provides sufficient detail while maintaining computational efficiency. A training subset percentage of 30% optimally balances training data, ensuring effective and not computationally costly learning. The number of keypoints was set to 30 for SIFT and 20 for ORB, maximizing feature extraction for SIFT while allowing ORB to remain efficient with fewer keypoints. These parameters significantly improved the classifiers' performance, as evidenced by better mAP scores, demonstrating the importance of hyperparameter tuning for image classification.

4 Conclusion and Future Improvements

In this study, the effectiveness of traditional image classification techniques using a Bag of Visual Words approach was explored, focusing on feature extraction methods (SIFT and ORB), K-Means clustering for visual vocabulary construction, and SVM classification. The results indicated that SIFT outperformed ORB, demonstrating higher mAP scores and better handling of distinct classes like 'Automobile.' This finding highlights the importance of selecting robust features for effective classification [1]. A key limitation of this study is the low resolution of the CIFAR-10 dataset, which comprises images sized at just 32x32 pixels. This resolution constraint may prevent the classifiers from capturing details essential for accurately distinguishing between visually similar classes.

Future improvements could involve exploring more advanced feature extraction techniques (such as CNNs), which have shown promising results in various computer vision tasks [1]. Additionally, augmenting the dataset could enhance the model's robustness, particularly for classes with less distinct features. Hyperparameter tuning could also be further optimized by exploring additional parameters and conducting more extensive and computationally costly grid searches. Incorporating ensemble learning techniques could potentially improve classification accuracy by combining the strengths of multiple models. Overall, this project lays the groundwork for further research and development in image classification methodologies.

References

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